

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

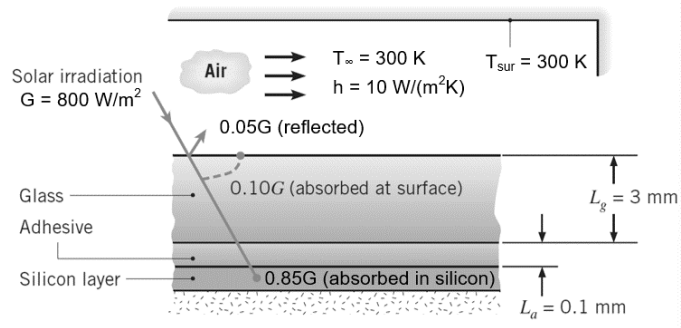
Spring 2026

Problem Set #4

Posted: Monday, Feb 23

Due: Tuesday, Mar 3, 11:59 pm

1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass ($k_g = 1.5 \text{ W/m}\cdot\text{K}$), a 0.1-mm-thick optical grade adhesive ($k_a = 1 \text{ W/m}\cdot\text{K}$), and a very thin layer of silicon within which solar energy is converted to



electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si} , as described by the expression $\eta = a - bT_{Si}$, where $a = 0.553$ and $b = 0.001 \text{ K}^{-1}$. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4} \text{ m}^2\cdot\text{K/W}$. There is no contact resistance between the glass and the adhesive.

In a sunny day, there is $G = 800 \text{ W/m}^2$ solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of $T_\infty = 300 \text{ K}$ and convective heat transfer coefficient of $h = 10 \text{ W/m}^2\cdot\text{K}$. The surrounding temperature is $T_{sur} = T_\infty = 300 \text{ K}$ and the effective radiative heat transfer coefficient is $h_{rad} = 5 \text{ W/m}^2\cdot\text{K}$. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

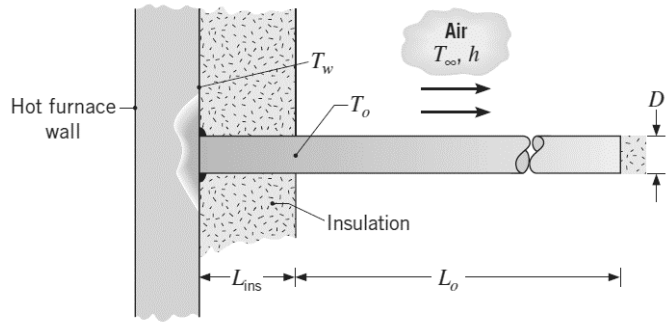
- (a) Show the thermal resistance network that can describe heat transfer across the PV panel.

Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the

temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

- (b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si} .
- (c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

2. (30 pt) A rod of diameter $D = 10$ mm and thermal conductivity $k = 80$ W/m·K protrudes normally from a furnace wall that is at $T_w = 200$ °C and is covered by insulation of thickness $L_{ins} = 200$ mm. The rod is welded to the furnace wall and is used as a hanger for supporting



instrumentation cables. The rod has an exposed length L_o and its tip is well insulated. To avoid damaging the cables, the temperature of the rod at its exposed surface, T_o , must be maintained below a specified operating limit of $T_{max} = 100$ °C. The ambient air temperature is $T_{\infty} = 20$ °C and convective heat transfer coefficient is $h = 20$ W/m²·K.

- (a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.
 - (b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o . Does the exposed length meet the specified operating limit?
 - (c) Calculate the minimum exposed length L_o that ensures T_o below the specified operating limit.
3. (30 pt) A square pin fin of uniform cross-section area is fabricated of copper ($k = 400$ W/m·K). The fin width is $s = 10$ mm and the fin is exposed to air flow with convective heat transfer coefficient $h = 100$ W/m²·K. It is known that the fin efficiency is $\eta_f = 0.6$. The fin has a convection tip.
- (a) Calculate the fin length L .
 - (b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f .

Problem #1

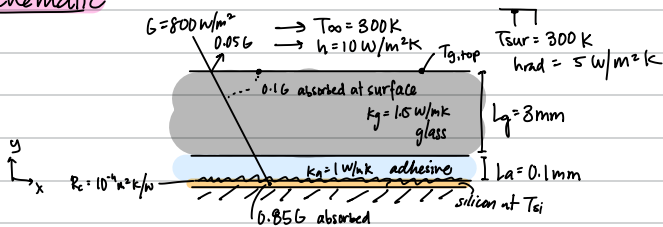
Known: $t_g = 3\text{mm}$, $t_a = 0.1\text{mm}$, $\eta = 0.553 - (0.001\text{K}^{-1})T_{\text{Si}}$, $R_c = 10^{-4}\text{m}^2\text{K/W}$
 $G = 800\text{W/m}^2$, 0.056 reflected, 0.16 absorbed at surface, 0.856 absorbed in silicon, $T_{\infty} = 300\text{K} = T_{\text{sur}}$

Find: a) thermal resistance network across PV panel

b) $T_{g,\text{top}}$, T_{Si}

c) η and power flux produced by PV panel

Schematic:

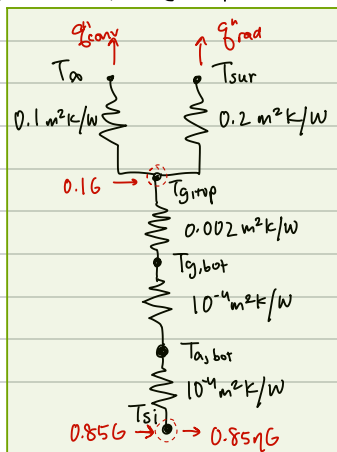


Properties: $k_g = 1.5\text{W/mK}$, $k_a = 1\text{W/mK}$, $h = 10\text{W/m}^2\text{K}$, $h_{\text{rad}} = 5\text{W/m}^2\text{K}$

Assumptions: negligible silicon thickness, perfectly thermally insulated, no contact resistance between glass & adhesive, uniform constant material properties & T_{sur} , no volumetric heat generation, 1D heat transfer, steady state

Analysis: a) There are 5 thermal resistances:

- 1) $R_c = 10^{-4}\text{m}^2\text{K/W}$
 - 2) $R_{a,\text{cond}} = \frac{L_a}{k_a} = (0.1 \times 10^{-3}\text{m})(1\text{W/mK})^{-1} = 1 \times 10^{-4}\text{m}^2\text{K/W}$
 - 3) $R_{g,\text{cond}} = \frac{L_g}{k_g} = (3 \times 10^{-3}\text{m})(1.5\text{W/mK})^{-1} = 0.002\text{m}^2\text{K/W}$
 - 4) $R_{\text{rad}} = \frac{1}{h_{\text{rad}}} = (5\text{W/m}^2\text{K})^{-1} = 0.2\text{m}^2\text{K/W}$
 - 5) $R_{\text{conv}} = \frac{1}{h} = (10\text{W/m}^2\text{K})^{-1} = 0.1\text{m}^2\text{K/W}$
- Resistances 2, 3, and 4 are in series. Resistances 1, 5, and the series combination of 2, 3, and 4 are in parallel.



Key assumptions: 1D heat transfer

- no volumetric heat generation
- steady state
- adiabatic base
- negligibly thin silicon

* Referenced example 3.3 in Fundamentals of Heat & Mass Transfer

b) Conservation of energy on node $T_{g, \text{top}}$ gives:

$$0.16 + \dot{q}_{g, \text{cond}} - \dot{q}_{\text{conv}} - \dot{q}_{\text{rad}} = 0$$

$$\text{Let } R_{\text{int}}^{\text{h}} = R_{\text{c}}^{\text{h}} + R_{\text{a, cond}}^{\text{h}} + R_{\text{g, cond}}^{\text{h}} = (10^{-4} + 10^{-4} + 0.002) \text{ m}^2 \text{K/W} = 0.0022 \text{ m}^2 \text{K/W}$$

$$R_{\text{env}}^{\text{h}} = \left(\frac{1}{R_{\text{conv}}} + \frac{1}{R_{\text{rad}}} \right)^{-1} = \left(\frac{1}{0.2} + \frac{1}{0.1} \text{ W/m}^2 \text{K} \right)^{-1} = \frac{1}{15} \text{ m}^2 \text{K/W}$$

Since $\dot{q}^{\text{h}} = \frac{\Delta T^{\text{h}}}{R^{\text{h}}}$ (and $T_{\text{surr}} = T_{\infty}$):

$$0.16 + \frac{T_{\text{si}} - T_{g, \text{top}}}{R_{\text{int}}^{\text{h}}} - \frac{T_{g, \text{top}} - T_{\infty}}{R_{\text{env}}^{\text{h}}} = 0$$

$$0.16 + \frac{T_{\text{si}}}{R_{\text{int}}^{\text{h}}} - T_{g, \text{top}} \left(\frac{1}{R_{\text{int}}^{\text{h}}} + \frac{1}{R_{\text{env}}^{\text{h}}} \right) + \frac{T_{\infty}}{R_{\text{env}}^{\text{h}}} = 0$$

Conservation of energy on node T_{si} gives:

$$0.856 - 0.856G - \frac{T_{\text{si}} - T_{g, \text{top}}}{R_{\text{int}}^{\text{h}}} = 0.85 - 0.85(a - bT_{\text{si}})G - \frac{T_{\text{si}} - T_{g, \text{top}}}{R_{\text{int}}^{\text{h}}} = 0$$

Solving for T_{si} in terms of $T_{g, \text{top}}$:

$$\left[0.856 - 0.856aG + 0.856G T_{\text{si}} - \frac{T_{\text{si}}}{R_{\text{int}}^{\text{h}}} + \frac{T_{g, \text{top}}}{R_{\text{int}}^{\text{h}}} = 0 \right] R_{\text{int}}^{\text{h}}$$

$$0.856 R_{\text{int}}^{\text{h}} (1-a) + 0.856 G R_{\text{int}}^{\text{h}} T_{\text{si}} - T_{\text{si}} + T_{g, \text{top}} = 0$$

$$T_{\text{si}} = \frac{T_{g, \text{top}} + 0.856 R_{\text{int}}^{\text{h}} (1-a)}{1 - 0.856 G R_{\text{int}}^{\text{h}}}$$

plug into first energy balance:

$$0.16 - T_{g, \text{top}} \left(\frac{1}{R_{\text{int}}^{\text{h}}} + \frac{1}{R_{\text{env}}^{\text{h}}} \right) + \frac{1}{R_{\text{int}}^{\text{h}}} \left(\frac{T_{g, \text{top}} + 0.856 R_{\text{int}}^{\text{h}} (1-a)}{1 - 0.856 G R_{\text{int}}^{\text{h}}} \right) + \frac{T_{\infty}}{R_{\text{env}}^{\text{h}}} = 0$$

$$0.16 + \left(0.856(1-a) \left(\frac{1}{1 - 0.856 G R_{\text{int}}^{\text{h}}} \right) + \frac{T_{\infty}}{R_{\text{env}}^{\text{h}}} \right) = T_{g, \text{top}} \left(\frac{1}{R_{\text{int}}^{\text{h}}} + \frac{1}{R_{\text{env}}^{\text{h}}} \right) - T_{g, \text{top}} \left(\frac{0.856 R_{\text{int}}^{\text{h}}}{1 - 0.856 G R_{\text{int}}^{\text{h}}} \right)$$

$$T_{g, \text{top}} = \frac{0.16 + 0.856(1-a) \left(\frac{1}{1 - 0.856 G R_{\text{int}}^{\text{h}}} \right) + \frac{T_{\infty}}{R_{\text{env}}^{\text{h}}}}{\left(\frac{1}{R_{\text{int}}^{\text{h}}} + \frac{1}{R_{\text{env}}^{\text{h}}} \right) - \left(\frac{0.856 R_{\text{int}}^{\text{h}}}{1 - 0.856 G R_{\text{int}}^{\text{h}}} \right)}$$

$$T_{g, \text{top}} = \frac{0.1(800 \text{ W/m}^2) + 0.85(800 \text{ W/m}^2)(1 - 0.553) \left(\frac{1}{0.0022} - 0.85(0.001 \text{ K}^{-1})(800 \text{ W/m}^2)(0.0022 \text{ m}^2 \text{K/W}) \right) + 300 \text{ K} \left(\frac{1}{15 \text{ W/m}^2 \text{K}} \right)}{\left(\frac{1}{0.0022} + 15 \right) \text{ W/m}^2 \text{K} - \left(\frac{0.0022}{1 - 0.85(0.001 \text{ K}^{-1})(800 \text{ W/m}^2)(0.0022 \text{ m}^2 \text{K})^2} \right)^{-1}}$$

$$T_{g, \text{top}} = 341.115 \text{ K}$$

$$T_{\text{si}} = \frac{341.115 \text{ K} + 0.85(800 \text{ W/m}^2)(0.0022 \text{ m}^2 \text{K/W})(1 - 0.553)}{1 - 0.85(0.001 \text{ K}^{-1})(800 \text{ W/m}^2)(0.0022 \text{ m}^2 \text{K/W})}$$

$$T_{\text{si}} = 342.296 \text{ K}$$

c) Since $T_{si} = 342.296 \text{ K}$:

$$\eta = a - b T_{si} = 0.553 - (0.001 \text{ K}^{-1} \cdot 342.296 \text{ K})$$
$$= 0.211 = 21.1\%$$

power flux:

$$P^* = 0.85 \eta G = 0.85 (0.211) (800 \text{ W/m}^2)$$
$$= 143.48 \text{ W/m}^2$$

Problem #2

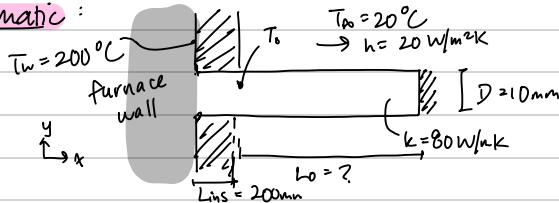
Known: $D = 10 \text{ mm}$, $T_w = 200^\circ\text{C}$, $L_{\text{ins}} = 200 \text{ mm}$, $T_{\text{max}} = 100^\circ\text{C}$, $T_{\infty} = 20^\circ\text{C}$

Find: a) expression for T_o (exposed surface temp)

b) T_o if $L_o = 200 \text{ mm}$

c) $L_{o,\text{min}}$ to ensure $T_o < T_{\text{max}}$

Schematic:



Properties: $k = 80 \text{ W/m}\cdot\text{K}$, $h = 20 \text{ W/m}^2\cdot\text{K}$

Assumptions: perfect insulation, steady state, constant properties, neglect radiation, 1D conduction, adiabatic tip, perfect thermal contact b/w wall & fin

Analysis: a) Since there is no volumetric heat generation and heat transfer is 1D, treat this as a thermal resistance problem with 2 resistances:

Referencing lecture 8:

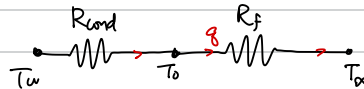
$$R_f = \frac{1}{\eta_f h A_f}$$

for a fin with an adiabatic tip: $\eta_f = \frac{\tanh mL}{mL}$

$$R_f = \frac{m L_o}{\tanh(m L_o) h A_f} = \frac{m L_o}{\tanh(m L_o) P h \cdot h} = \frac{\sqrt{\frac{h P}{k A_c}} L_o}{\tanh(m L_o) P h} = \frac{\sqrt{\frac{k A_c}{P h}}}{\tanh(m L_o)} = \frac{1}{\sqrt{k A_c P h} \tanh(m L_o)}$$

$$m = \sqrt{\frac{h P}{k A_c}} \quad \text{Table 3.4}$$

$$R_{\text{cond}} = \frac{L_{\text{ins}}}{k A_c}$$



$$\Delta T = q R$$

Energy balance on node T_o : $\frac{T_w - T_o}{R_{\text{cond}}} - \frac{T_o - T_{\infty}}{R_f} = 0$

$$\frac{T_w - T_o}{R_{\text{cond}}} = \frac{T_o - T_{\infty}}{R_f} \rightarrow \frac{R_f}{R_{\text{cond}}} (T_w - T_o) + T_{\infty} = T_o \rightarrow \frac{R_f}{R_{\text{cond}}} T_w + T_{\infty} = T_o + \frac{R_f}{R_{\text{cond}}} T_o$$

$$\frac{R_f}{R_{\text{cond}}} T_w + T_{\infty} = T_o \left(1 + \frac{R_f}{R_{\text{cond}}} \right)$$

$$\frac{R_f}{R_{\text{cond}}} = \frac{1}{\sqrt{k A_c P h} \tanh(m L_o)} \cdot \frac{k A_c}{L_{\text{ins}}} = \frac{\sqrt{\frac{k A_c}{P h}}}{L_{\text{ins}} \tanh(m L_o)} = \frac{\sqrt{\frac{k A_c}{P h}}}{L_{\text{ins}} \tanh(m L_o)} \cdot \frac{1}{m L_{\text{ins}} \tanh(m L_o)} = \frac{1}{m L_{\text{ins}} \tanh(m L_o)}$$

$$\rightarrow T_o = \frac{[m L_{\text{ins}} \tanh(m L_o)]^{-1} T_w + T_{\infty}}{1 + [m L_{\text{ins}} \tanh(m L_o)]^{-1}} \quad \text{with } m = \sqrt{\frac{h \cdot \pi D}{k \cdot \pi D^2}} = \sqrt{\frac{4h}{kD}}$$

b) Given $L_0 = 200 \text{ mm}$:

$$T_0 = \frac{[m L_0 \tanh(m L_0)]^{-1} T_w + T_{\infty}}{1 + [m L_0 \tanh(m L_0)]^{-1}}$$

$$m = \sqrt{\frac{4h}{kD}} = \sqrt{\frac{4(20 \text{ W/m}^2\text{K})}{(80 \text{ W/mK})(0.01 \text{ m})}} = 10 \text{ m}^{-1}$$

$$T_0 = \frac{[10 \text{ m}^{-1} (0.2 \text{ m}) \tanh(10 \text{ m}^{-1} \cdot 0.2 \text{ m})]^{-1} \cdot (200^\circ\text{C}) + 20^\circ\text{C}}{1 + [10 \text{ m}^{-1} (0.2 \text{ m}) \tanh(10 \text{ m}^{-1} \cdot 0.2 \text{ m})]^{-1}}$$

$$T_0 = 81.47^\circ\text{C}$$

Since $81.47^\circ\text{C} < 100^\circ\text{C}$, the exposed length meets the specified operating limit.

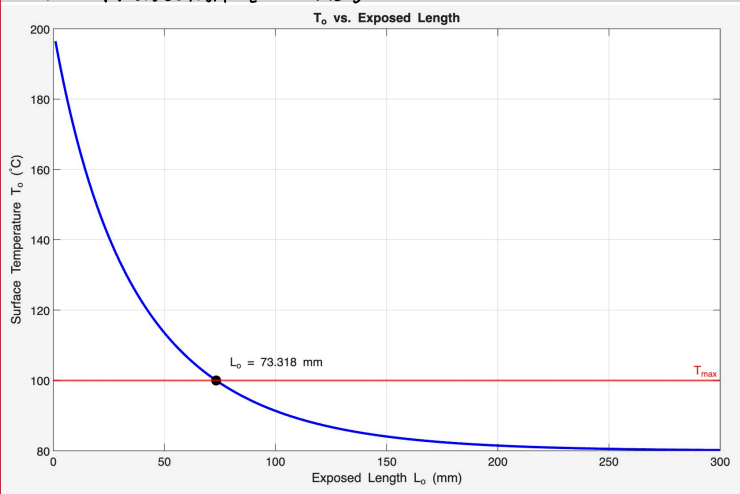
c) Set T_0 equal to $T_{\max}$ to find $L_{0,\min}$:

$$T_{\max} = \frac{[m L_{\min} \tanh(m L_{\min})]^{-1} T_w + T_{\infty}}{1 + [m L_{\min} \tanh(m L_{\min})]^{-1}}$$

Plot $f = 100^\circ\text{C}$

$$\text{and } g(L_{\min}) = \frac{[10 \text{ m}^{-1} (0.2 \text{ m}) \tanh(10 \text{ m}^{-1} \cdot L_{\min})]^{-1} \cdot (200^\circ\text{C}) + 20^\circ\text{C}}{1 + [10 \text{ m}^{-1} (0.2 \text{ m}) \tanh(10 \text{ m}^{-1} \cdot L_{\min})]^{-1}}$$

→ find intersection (MATLAB):



$$L_{0,\min} = 73.318 \text{ mm}$$

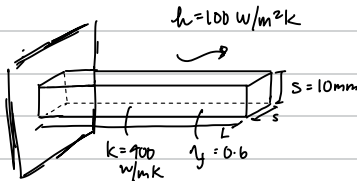
Problem #3

Known: $s = 10 \text{ mm}$, $\eta_f = 0.6$, convection tip

Find: a) L , length of fin

b) R_f and ϵ_f

Schematic



Properties: $k = 400 \text{ W/mK}$, $h = 100 \text{ W/m}^2\text{K}$

Assumptions: uniform properties, 1D heat conduction, steady state, neglect tip area in A_f

Analysis: a) According to page 155 of the textbook, with tip convection, the heat rate can be approximated as $q_f = M \tanh m L_c$, where $L_c = L + \frac{s}{2}$

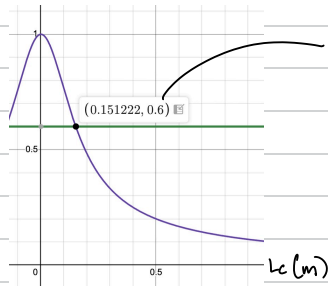
This approximation has negligible errors only when $hs/k \leq 0.0625$, so first check validity: $\frac{hs}{k} = \frac{(100 \text{ W/m}^2\text{K})(0.01 \text{ m})}{400 \text{ W/mK}} = 0.0025 < 0.0625$
 $\therefore$ The approximation holds.

This gives: $\eta_f = \frac{\tanh m L_c}{m L_c}$ (eqn. 3.95)

$$m = \sqrt{\frac{hP}{KA_c}} = \sqrt{\frac{h4\delta}{ks^2}} = \sqrt{\frac{4h}{ks}} = \sqrt{\frac{4(100 \text{ W/m}^2\text{K})}{400 \text{ W/mK} \cdot 0.01 \text{ m}}} = 10 \text{ m}^{-1}$$

knowing $\eta_f = 0.6$:

$$0.6 = \frac{\tanh(10 L_c)}{10 L_c}$$



This equation is satisfied when $L_c = 0.1512 \text{ m}$ or 151.22 mm

$$\text{Since } L_c = L + \frac{s}{2} \rightarrow L = L_c - \frac{s}{2} = 151.22 \text{ mm} - \frac{10 \text{ mm}}{2} = 146.22 \text{ mm}$$

$$b) R_f = \frac{1}{\eta_f h A_f} \quad (\text{eqn. 3.93})$$

$$= \frac{1}{\eta_f h A_f} = (0.6 \cdot 100 \text{ W/m}^2\text{K} \cdot 4 \cdot 0.01 \text{ m} \cdot 0.14622 \text{ m})^{-1}$$

$$= 2.85 \text{ K/W}$$

$$E_f = \frac{R_b}{R_f} \quad (\text{eqn. 3.90})$$

$$= \frac{1}{k A_b} \cdot \eta_f h A_f = \frac{\eta_f A_f}{A_b} = \frac{\eta_f \cancel{48} L}{\cancel{8}^2} = \frac{\eta_f L}{8} = \frac{1(0.6)(0.14622 \text{ m})}{0.01 \text{ m}} = 35.09$$